Burnaby, BC
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Portfolio

Erfan Shafagh

Machine Learning Researcher

erfan-shafagh
erfanshafagh

A computer science graduate student at Simon Fraser University specializing in machine learning, with research experience in developing, fine-tuning, and benchmarking deep learning and large language models on real-world datasets. Strong technical background in Python, PyTorch, model training pipelines, data preprocessing, hyperparameter tuning, and quantitative evaluation. Seeking opportunities to contribute to impactful academic research through strong technical depth, rigorous experimentation, and collaborative problem-solving.

Skills

Languages	Python, C++, C, Java
Frameworks / Libraries	PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face
Databases	SQL, PostgreSQL, SQLite
Developer Tools	Linux, Git, GitHub, Docker, VS Code, Google Colab
Fundamentals	Data Structures and Algorithms, Object-Oriented Programming, Database Design
ML Techniques	Semi-supervised Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning
Soft Skills	Team Collaboration, Clear Communication, Reliability, Accountability, Agile Methodology

Education

Master of Computer Science | Simon Fraser University — **May 2026 - Present**

- CGPA: **4.0**
- Research Areas: LLM-based vulnerability detection, commit-level vulnerability detection, automated vulnerability patching, and agentic graph-based program analysis.

Bachelor of Computer Science | Simon Fraser University — **Jan 2023 - Dec 2025** (3 years)

- CGPA: **3.46** | Upper Division GPA: **3.7**
- Top Coursework: Deep Learning, Computer Vision, Computer Simulation, Natural Language Processing, Geometric Modelling, Machine Learning, Computational Data Science, Database Systems.

Experience

Research Assistant | Simon Fraser University | TBLab — **Nov 2024 - Present**

- Gained hands-on experience with the full research cycle, from problem formulation to experimentation and evaluation.
- Collaborated in an academic research setting, contributing to publications and interdisciplinary teamwork.
- Conducted research in fraud detection techniques using weak supervision and anomaly detection methods.
- Applied deep learning methods to real-world, high-dimensional datasets.

AI Engineer Intern | Farpoint Technologies — **May 2026 - Present**

- Contributed to Fabric, a secure and private AI assistant built as an agentic harness around large language models.
- Designed and shipped core product features, including streaming reasoning components, onboarding flows, and file search.
- Diagnosed and resolved cross-platform bugs spanning encoding, rendering, and WSL environments.

Teaching Assistant | Simon Fraser University | CMPT 125 — **May 2026 - Present**

- Led lab sections and office hours, guiding students through core programming concepts, data structures, and algorithm design in C.
- Graded assignments and exams, providing detailed feedback to help students strengthen their problem-solving skills.

Program Coordinator | FOSINT-SI Symposium — **Mar 2026 - Present**

- Managed paper submissions and author–reviewer communication.
- Designed and maintained the conference website and media content.

Publications

CleverCatch: A Knowledge-Guided Weak Supervision Model for Fraud Detection — **2025**
A. Mozafari, K. Hashemi, **E. Shafagh**, S. Motamedi, A. Taheri, and M. A. Tayebi — **IEEE BigData**

Machine Learning Projects

Agent-Based Adversarial & Defensive Learning for Web Application Firewalls | GitHub — **Jul - Dec 2025**

- Researched and implemented adversarial attack generation techniques (e.g., SQLi, XSS) using language models.
- Fine-tuned multiple LLMs using reinforcement learning with GRPO, optimizing them to bypass WAFs.
- Designed and tested custom reward functions to improve the effectiveness and diversity of generated attack payloads.

AI Song Lyric Generation — Feb - Apr 2025

- Fine-tuned a transformer-based model (GPT-2), to generate genre-specific (pop, rap) song lyrics.
- Engineered the end-to-end training and inference pipeline utilizing PyTorch and the Hugging Face Transformers library.
- Compared parameter-efficient fine-tuning techniques, including Residual Adapters and LoRA, to adapt the LLM effectively.
- Evaluated model performance using perplexity metrics and human assessment of lyrical quality and genre relevance.

Aerial Image Instance Segmentation — Jan - Apr 2025

- Built an instance segmentation pipeline for high-resolution aerial imagery using PyTorch and Detectron2.
- Handled large-scale data by implementing custom data loaders, block-based image tiling, and augmentations.
- Fine-tuned a Faster R-CNN model to accurately localize aerial objects and maximize Average Precision.
- Engineered a custom encoder-decoder neural network with skip connections for precise pixel-level masking.
- Optimized segmentation performance using Intersection-over-Union (IoU) metrics and custom loss functions.

Optical Character Recognition System — Jan - Feb 2025

- Implemented a Convolutional Neural Network from scratch using NumPy and writing custom forward and backward passes.
- Developed an Optical Character Recognition (OCR) pipeline using OpenCV to segment, extract, and classify handwritten digits.
- Trained and evaluated the model on the MNIST dataset, achieving over 95% test accuracy.
- Visualized intermediate network feature maps to evaluate how effectively the convolutional and ReLU layers extracted edges.

Movie Title Generator | GitHub — Jul 2024

- Built a generative Multilayer Perceptron (MLP) from the ground up to produce creative, novel movie titles learned purely from data.
- Trained the model in PyTorch with CUDA GPU acceleration, gaining hands-on intuition for the full deep learning training loop.
- Preprocessed a real-world movie title dataset, handling formatting to prepare sequences for character-level generation.

Software & Systems Projects

Corridor Robot | MARS Lab | GitHub — Sep - Nov 2025

- Developed a real-time path planning system in ROS2 for an RB1-base robot to navigate corridors while avoiding humans.
- Implemented Model Predictive Path Integral (MPPI) control for smooth and safe motion planning in real time.
- Utilized JAX for efficient parallel computation, enabling fast real-time trajectory optimization.
- Optimized the MPPI cost function to improve human avoidance, coverage efficiency, and trajectory smoothness.

Autonomous Cleaning Robot Path Planning | MARS Lab | GitHub — Jan - Apr 2025

- Developed a path planning algorithm using ROS2 for a TurtleBot3 robot to autonomously clean rooms with static obstacles.
- Implemented a grid-based environment representation, dynamically updating cell status.
- Utilized Monte Carlo Tree Search to explore potential cleaning sequences, simulating actions and selecting the optimal path.
- Optimized the MCTS reward system for efficient coverage, collision avoidance, and path efficiency.

Data-Driven Urban Exploration Tools — Apr 2025

- Implemented a location-based service to find and filter nearby amenities using Haversine distance and user preferences.
- Created a tour planning feature that generates optimal walking, biking, or driving routes via OpenRouteService.
- Designed a virtual tour guide simulating exploration by identifying nearby landmarks along interpolated paths.
- Utilized pandas and numpy for efficient data manipulation and computation throughout the pipeline.

Cinepass — Sep - Nov 2024

- Engineered a full-stack web application within a 6-person team to manage movie ticketing and reservations.
- Developed the backend using Java Spring Boot to handle dynamic movie listings and event bookings.
- Implemented secure user authentication and robust relational data storage using PostgreSQL.
- Accelerated project delivery through Agile methodologies and iterative sprint cycles.

Socket Networking Chat Program | GitHub — Feb 2024

- Engaged with a team to use UDP sockets in C to achieve two-way communication between users over a network.
- Handled simultaneous operations by utilizing a multithreaded approach to enhance the responsiveness of the chat program.
- Ensured efficient system communication by using UDP sockets to implement a reliable messaging system.

Spotify Playlist Automation | GitHub — Jun 2023

- Built a Python program that scans local audio libraries, extracts metadata, and syncs tracks directly to a Spotify playlist.
- Integrated OAuth-based Spotify API authentication to enable secure, real-time playlist management programmatically.
- Engineered multi-format audio parsing to support MP3, FLAC, and M4A files using format-specific metadata libraries.

Awards & Honors

Spring 2024	President’s Honour Roll, Simon Fraser University
Spring 2025 & Summer 2024	Dean’s Honour Roll, Simon Fraser University